


Designed by Remus Teodorescu 2026

LECTURE 10.2

# Electrolysers

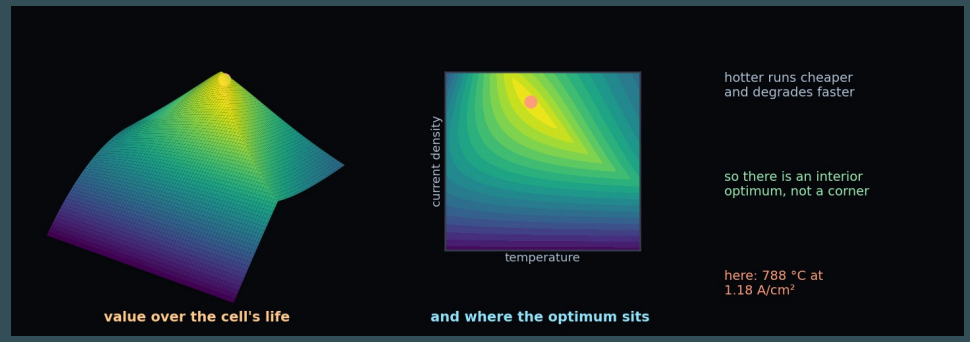

**Remus Teodorescu**

*With support from Research Assistant Noman Khan - nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk


Deep Learning for Engineering · 2026

STEAM · VOLTAGE · LIFETIME




*Schematic: the value a cell produces over its lifetime against temperature and current, with its optimal point in the interior - hotter is cheaper per hour and ages faster. The lecture sets out the electrolyser and the thermoneutral voltage, solves the fuel channel, and finds the optimal voltage by a gradient through one PINN trained over the whole operating range.*

# Learning Objectives

## SECTIONS, IN ORDER

1. **The cell** — three electrolysers, the losses, the thermoneutral voltage

2. **The channel** — steam, current and heat settling together

3. **The optimal voltage** — one PINN for every operating point, the cost and its gradient

*Incropera carries the transport, Liu the optimisation through a network. The example is Ex_10.2's electrolysis cell: the channel of L9.2, now with its electrode. This is the last physics lecture, and the place Part 2 has been heading.*

## READ, AND WHAT EACH ANSWERS

| | |
|---|---|
| **Incropera ch. 1** | where do the losses sit? |
| **Incropera ch. 14** | how does the steam run short? |
| **Liu ch. 5** | what does a gradient buy? |

# Three Electrolysers

**Water electrolyser** — a cell that splits water into hydrogen and oxygen with electricity. The three industrial types differ in their electrolyte, and so in the ion that carries the charge and in the temperature at which they run.

## ONE ELECTROLYTE EACH

**alkaline** — a potassium hydroxide solution near 80 degrees; $OH^-$ carries the charge and a diaphragm keeps the gases apart

**PEM** — a humid polymer membrane near 70 degrees; $H^+$ crosses to the hydrogen side and liquid water is split at the oxygen side

**solid oxide** — a ceramic near 800 degrees; $O^{2-}$ crosses to the air side, steam is split and heat supplies part of the energy

## COST AND CONTROLLABILITY

| | alkaline | PEM | solid oxide |
|---|---|---|---|
| temperature [°C] | 70 – 90 | 50 – 80 | 700 – 850 |
| energy [kWh/kg $H_2$] | 50 – 78 | 50 – 83 | 45 – 55 |
| cost [USD/kW] | 500 – 1000 | 700 – 1400 | stack >2000 |
| minimum load [%] | 10 – 30 | 10 – 40 | about 50 |
| ramp [% per s] | 0.5 – 5 | 3 – 10 | about 0.2 |
| start from cold | minutes | minutes | hours: kept hot |

*Typical values, rounded; costs are IRENA's 2020 system prices (installed plants more: 2000 – 2600 USD/kW outside China, 600 – 1200 in China; IEA 2025). The solid oxide rows are one product's data sheet, and its limits are thermal, not electrical: the stack follows the current within seconds, and designs that manage the heat, such as Dynelectro's AC:DC cycling, reach low minimum load.*



*IRENA (2020); Dorn et al. (2026); IEA (2025)*

# Electrolysis at Industrial Scale

**Power-to-X** — the conversion of renewable electricity into hydrogen by electrolysis, and of the hydrogen into a fuel or chemical, such as methanol, that can be stored and shipped.

## KASSØ, DENMARK, SINCE 2025

| | |
|---|---|
| **electrolysers** | 52 MW of PEM, from Siemens Energy |
| **power** | a 304 MW solar park beside the plant |
| **product** | up to 42 000 t of e-methanol a year |

## INSTALLED WORLDWIDE, AND 2030

| | |
|---|---|
| **installed** | 0.6 GW in 2021, 2 GW in 2024, 4.2 GW in 2025 |
| **by technology** | in 2022: alkaline 60 %, PEM 30 %, SOEC under 1 % |
| **by 2030** | 26 GW committed; 520 GW announced, much now out of reach |
| **China** | 65 % of the capacity installed or committed |




Laura Mærsk, e-methanol fuel
Jens Dohrn, CC BY-SA 4.0




2 MW PEM module, ITM Power
Bubble60, CC BY-SA 4.0

## WHY ALKALINE STILL LEADS

| | |
|---|---|
| **mature** | a century in industry |
| **cheap** | no noble metals, large stacks |
| **China's makers** | alkaline at 600 – 1200 USD/kW |

*Elsewhere 2000 – 2600 USD/kW (IEA). More: Kassø, ITM Power.*

# The Solid Oxide Electrolyser

**Solid oxide electrolysis cell** — a ceramic electrochemical cell that splits steam into hydrogen and oxygen near 800 degrees. The name comes from its electrolyte, a solid oxide (yttria-stabilised zirconia) that conducts only $O^{2-}$ ions. In reverse it is a solid oxide fuel cell; the electrodes are therefore the fuel and air electrode, as anode and cathode swap between modes.

## WHY THE HIGH TEMPERATURE PAYS

$$\Delta H = \Delta G + T\Delta S, \qquad \Delta G \downarrow \text{ as } T \uparrow$$

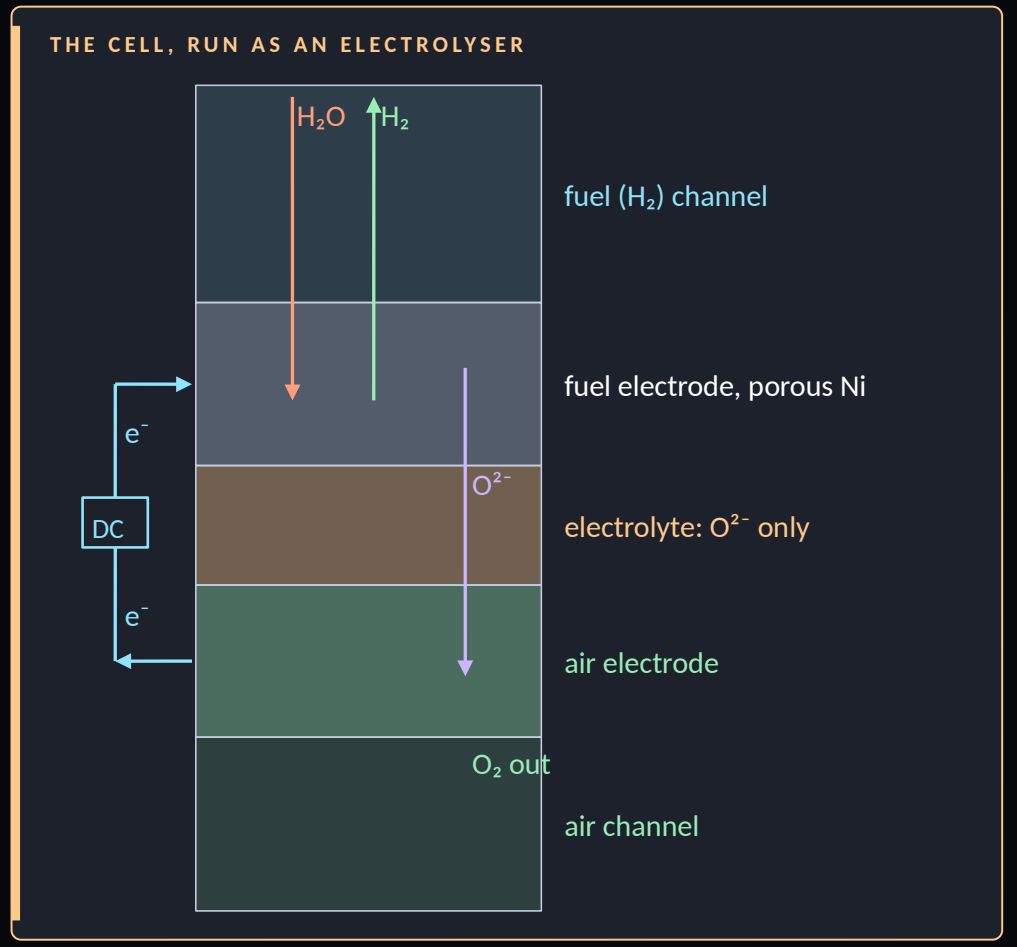
$\Delta H$ the energy to split water, $\Delta G$ the part that must be electricity, $T\Delta S$ the part heat can supply

## THE THREE LAYERS, AND THEIR REACTIONS

**fuel electrode** — $H_2O + 2e^- \rightarrow H_2 + O^{2-}$: two electrons from the supply; the oxide ion enters the ceramic

**electrolyte** — $O^{2-}$ hops through oxygen vacancies in the hot ceramic; electrons and gases cannot pass

**air electrode** — $2\,O^{2-} \rightarrow O_2 + 4e^-$: the ions return their electrons to the circuit and leave as oxygen



# The Polarisation Curve

**Nernst potential** — the voltage at which no current flows: the reversible voltage of the reaction, set by the local gas composition and the temperature; it rises as the steam is used up. The polarisation curve is this potential plus the losses, which grow with the current.

## THE NERNST POTENTIAL, AND THE LOSSES ON TOP

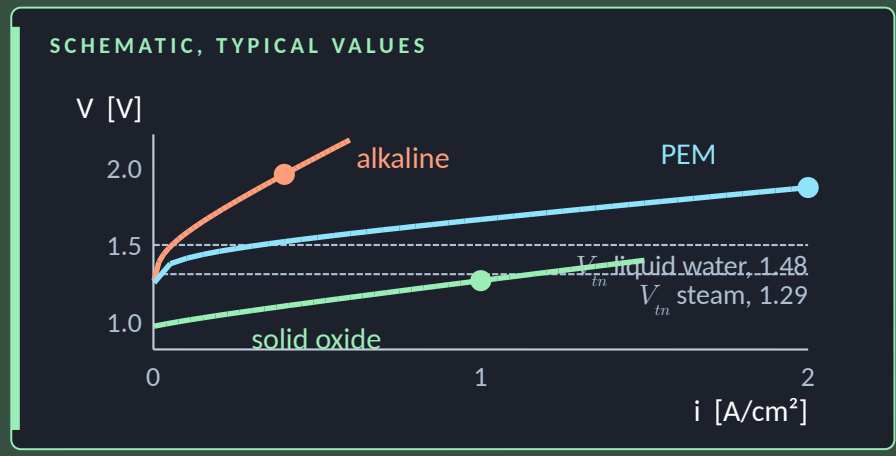
$$E = E^0(T) + \frac{RT}{2F}\ln\left(\frac{(1-y)\,p_{O_2}^{1/2}}{y}\right)$$

$$V = E + \eta_{act} + \eta_{ohm} + \eta_{conc}$$

$E$ the Nernst potential, $y$ the steam fraction, $p_{O_2}$ the air side's oxygen, $V$ the cell voltage
$\eta_{act}, \eta_{ohm}, \eta_{conc}$ the activation, ohmic and gas-transport losses; in electrolysis they add

Ex_10.2 keeps the losses separate: activation, a resistance that falls with temperature, and gas transport with its limiting current - every constant an estimate, flagged in the notebook.

## THREE CELLS: VOLTAGE TO ENERGY PER KG




| | |
|---|---|
| **alkaline** | 1.94 V at 0.4 A/cm²: 52 kWh/kg |
| **PEM** | 1.85 V at 2.0 A/cm²: 49 kWh/kg |
| **solid oxide** | 1.25 V at 1.0 A/cm²: 33 kWh/kg |

*One volt costs 26.6 kWh/kg (stack); a system needs 45 – 55.*

# The Thermoneutral Voltage

**Thermoneutral voltage** — the operating voltage at which the heat the losses release equals the heat the reaction absorbs, so the net heat is zero. For steam near 800 degrees it is 1.29 volts; stacks are operated at or near it, and with outside heat a stack can run below it, endothermically.

## THE BALANCE, AND THE HEAT IN ONE TERM

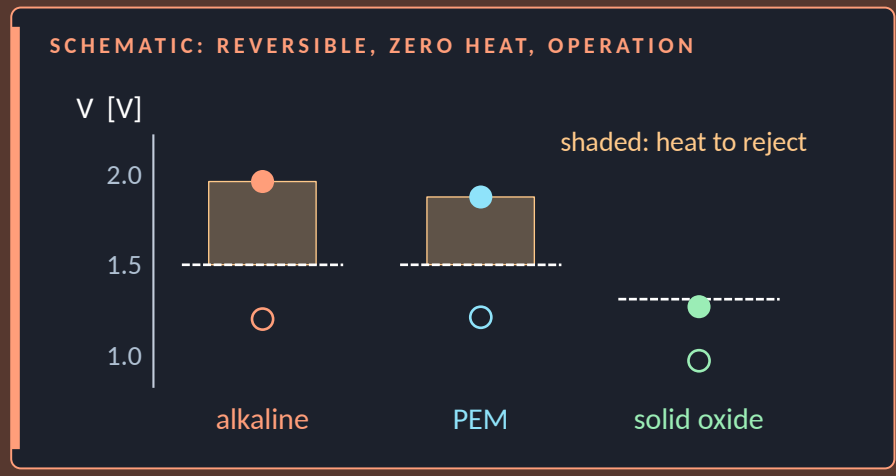
$$V_{tn} = \frac{\Delta H}{2F} = 1.29\ \text{V} \quad (\text{steam},\ 800\ ^\circ\text{C})$$

$$q = i(V - V_{tn})$$

$V_{tn}$ the thermoneutral voltage, $\Delta H$ the enthalpy of splitting, $q$ the heat per electrode area, $i$ the current density

**TALK · 2 MIN** How would you explain, in your own words, where the heat of an electrolyser comes from and where it goes? Then, in the drawing on the right, why do the two cold cells reject heat while the solid oxide cell runs at its dashed line?

## THREE CELLS ON ONE VOLTAGE AXIS




**alkaline, PEM** 0.4 – 0.5 V above: heat to remove

**solid oxide** at $V_{tn}$: no net heat in the stack

*Liquid water: 1.48 V includes evaporation. Steam at 800 °C: 1.29 V, the steam raised outside.*

# The Fuel Channel

**Fuel channel** — the hydrogen channel of L9.2, run as an electrolyser: 90 % steam enters at 1 m/s and the electrode takes steam and returns hydrogen, so the steam mole fraction $y$, the share of steam molecules in the gas, falls along the channel.

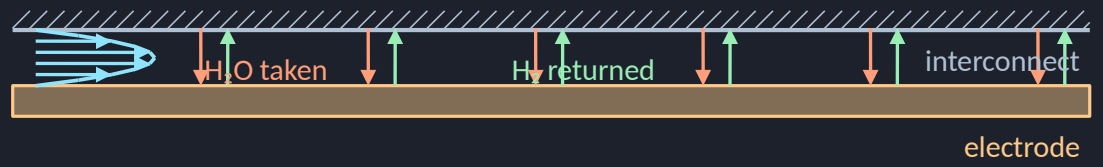
## THE TWO SCALED BALANCES OF EX_10.2

$$y_x = \frac{y_{xx}}{Pe} - K_I\, i, \qquad Pe = 125$$

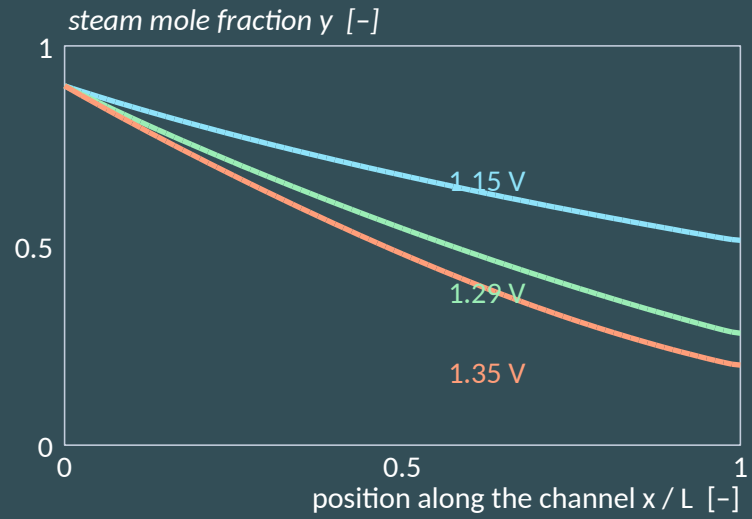
$$\theta_x = \frac{\theta_{xx}}{Pe_T} + S_Q\, i(V - V_{tn}), \qquad Pe_T = 16, \quad \theta = \frac{T - T_{in}}{50\ \mathrm{K}}$$

$y$ the steam mole fraction, $\theta$ the scaled temperature, $i$ the local current density [A/cm$^2$]
$Pe, Pe_T$ the Péclet numbers; $K_I, S_Q$ gather $u, h, c, F, \Gamma, L$; $V_{tn} = 1.29$ V

### STEAM TAKEN, HYDROGEN RETURNED




### STEAM ALONG THE CHANNEL




### THE CHANNEL IN THE THREE CELLS

| | fed | the channel must |
|---|---|---|
| alkaline | KOH solution, in excess | carry gas bubbles out |
| PEM | liquid water, in excess | carry bubbles, cool |
| solid oxide | steam, 69 % used | keep steam to the outlet |

# The Local Current

**Local current density** — the current per electrode area at one point. The cell is held at one voltage $V$; the local steam fraction and temperature set the Nernst potential, the kinetics and the resistance, and the current is whatever satisfies the charge transfer there.

## ONE VOLTAGE, A CURRENT THAT VARIES

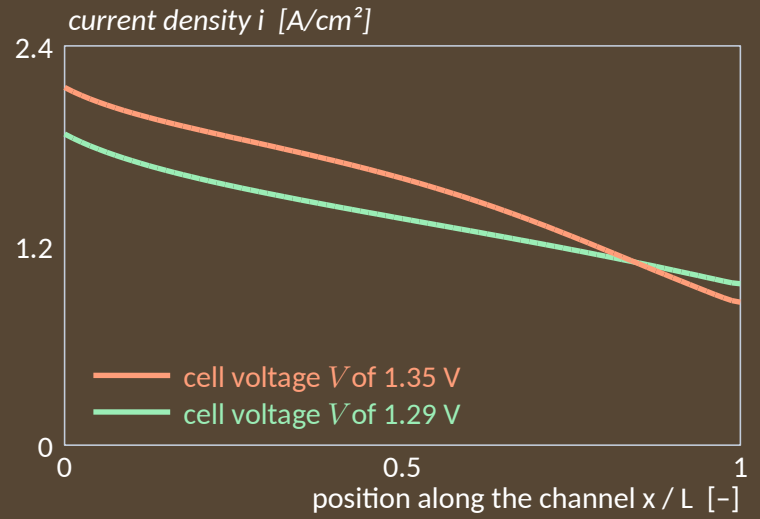
$$V = E(y,T) + \frac{RT}{F}\,\mathrm{asinh}\frac{i}{2i_0} + i\,ASR(T) - \frac{RT}{2F}\ln\left(1 - \frac{i}{i_L(y)}\right)$$

$E(y,T)$ the Nernst potential of the local gas, $i_0$ the exchange current density, $ASR(T)$ Arrhenius (estimates)
$i_L(y)$ the limiting current, proportional to $y$; no derivative: an equation for $i$ at every point

## WHY IT FALLS, AND WHAT IT MEANS

| | |
|---|---|
| **1 · steam used** | the steam fraction $y$ falls along the channel |
| **2 · $E$ rises** | the Nernst potential grows as $y$ falls |
| **3 · $i$ falls** | and so does the heat it makes or takes |
| **4 · $T$ answers** | the resistance follows the temperature |
| **→ multiphysics** | no field can be solved first and handed on |

## THE CURRENT ALONG THE CHANNEL




## THE CURRENT IN THE THREE CELLS

| | along the channel | set by |
|---|---|---|
| alkaline | nearly even | gas bubbles |
| PEM | nearly even | bubbles, membrane |
| solid oxide | 1.87 to 0.97 A/cm² | steam used up |

# Finite Volumes Along the Channel

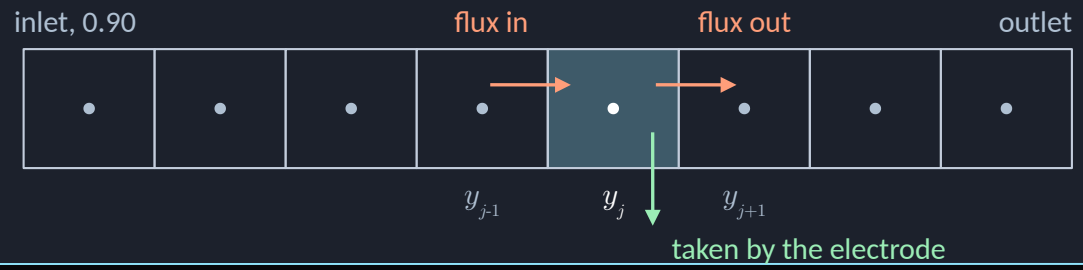
**Finite volumes (FVM)** — the channel cut into cells, and for each cell the balance of its steam and of its heat: what flow and diffusion carry in through one face, less what leaves by the other, equals what the electrode takes or the current releases. Shared faces conserve both exactly.

## THE BALANCE OF CELL j

$$\frac{y_j - y_{j-1}}{h} - \frac{y_{j+1} - 2y_j + y_{j-1}}{Pe\,h^2} + K_I\, i_j = 0, \qquad \text{and its twin for } \theta_j, \qquad j = 1, \ldots, n$$

$y_j, \theta_j$ steam and scaled temperature in cell $j$, $h = 1/n$, $i_j$ the root of the charge transfer in cell $j$
$2n$ equations, solved together by Newton-Raphson; the coupling blocks carry $\partial i/\partial y$ and $\partial i/\partial \theta$

### EIGHT OF THE n CELLS, CELL j PICKED OUT




## REFINED, AGAINST THE REFERENCE

| cells | steam | kelvin | |
|---|---|---|---|
| 25 | 0.0131 | 0.55 | |
| 50 | 0.0052 | 0.29 | meets 0.01, 0.5 K |
| 100 | 0.0024 | 0.14 | |
| 200 | 0.0015 | 0.07 | |

*The error falls about in proportion to the cell width, because the flow term takes the upstream cell. One solve is a handful of Newton-Raphson steps on the 2n unknowns with their two coupling blocks, some 25 ms, repeated for every operating point.*

# One PINN for Every Operating Point

**Parametric PINN** — a PINN trained over three domains at once: the position $x$ along the channel, the cell voltage $V$ from 1.15 to 1.35 V and the inlet temperature from 750 to 850 degrees. After this one training it returns all three fields at any operating point in that range.

## THE PHYSICS IN THE TRAINING

**steam balance** $$y_x = \frac{y_{xx}}{Pe} - K_I\, i$$

**energy balance** $$\theta_x = \frac{\theta_{xx}}{Pe_T} + S_Q\, i(V - V_{tn})$$

**charge transfer** $$V_{cell}(i, y, T) = V, \qquad i = i_L(y)\,\sigma(\mathcal{N}_3)$$

**boundaries** $$y = 0.90 + x\,\mathcal{N}_1, \quad \theta = x\,\mathcal{N}_2; \qquad y_x = \theta_x = 0 \text{ at } x = 1$$

$$\mathcal{L}(\theta) = \overline{r_y^2} + \overline{r_\theta^2} + \overline{\left(\frac{V_{cell} - V}{50\ \text{mV}}\right)^2} + \lambda\, \overline{\text{outlet slopes}^2}$$

$r_y, r_\theta$ the residuals of the two balances at points $(x, V, T_{in})$ over 1.15 to 1.35 V and 750 to 850 °C
the charge residual in units of 50 mV, so millivolts weigh like the balances; $\lambda$ the outlet-slope weight

## WHAT THE ONE TRAINING BUYS

| | |
|---|---|
| **one point** | FVM in 25 ms is the tool |
| **1000 points** | one batch, about 500 times faster |
| **then** | the cost $C(V)$ and $dC/dV$ |
| **accuracy** | 0.0022, 0.19 K; FVM 0.0068, 0.40 K |

### THE TRIAL FUNCTION IN PYTORCH

```
N1,N2,N3 = MLP(3,3,64,4)(p)   # x, V, T_in in
y = Y_IN + x*N1; th = x*N2    # inlets exact
i = i_lim(y)*sigmoid(N3)      # 0 <= i < i_L
```

*Trained once, offline: 33 min on a CPU, repaid after some 80 000 operating points.*

# Optimisation of the Cell Voltage

**Cost function** — the number the operator minimises: electricity bought, less hydrogen sold, plus the share of the stack aged in the hour, as a function of the cell voltage $V$. The prices and the ageing law are assumptions; the optimum is conditional on them.

## THE COST, ITS UNITS, AND ITS MINIMUM

$$C(V) = p_{el}\,\frac{V\,I}{1000} - p_{H_2}\,\frac{3600\,I\,M_{H_2}}{2F} + \frac{c_{stack}\,P_{rated}}{\Delta ASR_{EOL}}\,\Delta ASR_h$$

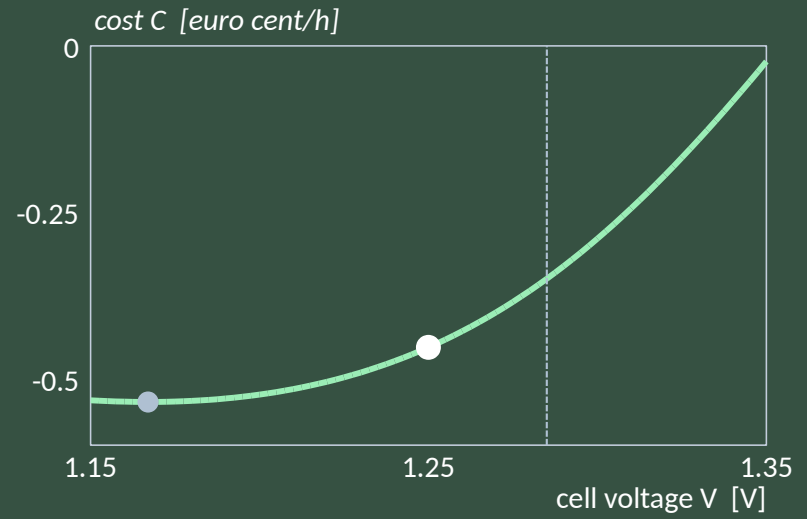
$$I = A\int_0^1 i\,dx \;\;[\mathrm{A}], \qquad \frac{V\,I}{1000}\;[\mathrm{kW}], \qquad \frac{3600\,I\,M_{H_2}}{2F}\;[\mathrm{kg/h}], \qquad V^* = \arg\min_V C \;\; \text{s. t.} \;\; \max_x |T - T_{in}| \le 10\ \mathrm{K}$$

$C$ [cent/h] (negative : a profit); $p_{el} = 0.05$ euro/kWh, $p_{H_2} = 5$ euro/kg : assumptions; the 1000 converts W to kW the 3600 converts per second to per hour; $\Delta ASR_h$ the resistance grown in the hour (the assumed law below)

### THE OPTIMAL VOLTAGE IN PYTORCH

```
V = torch.tensor(1.25, requires_grad=True)   # the decision variable
J = C(V) + mu * relu(dT(V) - 10.0)**2        # the cost, the limit as a penalty
J.backward(); V -= lr * V.grad               # downhill through the network
```

### THE COST AGAINST THE VOLTAGE




| | |
|---|---|
| **free optimum** | 1.167 V, but the cell cools 25 K |
| **with the limit** | $V^*$ = 1.250 V, 0.45 cent/h earned |
| **checked by** | a sweep of 81 solver runs |

# No Optimal Point Over a Lifetime

**Degradation** — the slow rise of the cell's area-specific resistance over thousands of hours, from changes in its layers (right). The rate law below is an empirical form; its constants in Ex_10.2 are assumed, and the grown resistance only prices the hour - it is not fed back into the model.

## THE DEGRADATION LAW, AND THE LIFETIME OBJECTIVE

$$\frac{dASR}{dt} = k_0 \exp\left(-\frac{E_d}{RT}\right) |i|^n$$

$$\min_{V(t),\,T(t)} \int_0^{t_{life}} [p_{el} P - p_{H_2} \dot{m}_{H_2}]\, dt \quad \text{s.t.} \quad ASR(t_{life}) \le ASR_{\max}, \quad |\nabla T| \le |\nabla T|_{\max}$$

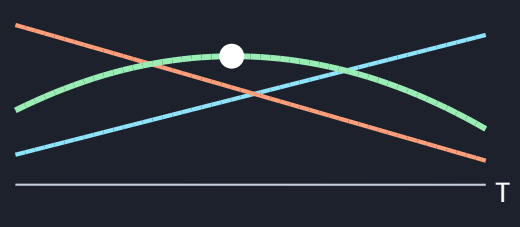
$k_0, E_d, n$ ASSUMED in Ex_10.2, of the right order (published rates span an order of magnitude) an empirical form, Arrhenius in $T$ and a power of $i$, not derived from the mechanisms on the right

**TALK · 2 MIN** How would you explain, in your own words, why a device like this has no best operating point? Then, in the drawing on the right, what moves the white dot when the price of electricity falls?

## WHAT AGES, AND WHY

| layer | mechanism |
|---|---|
| fuel electrode | nickel grains coarsen and migrate: fewer reaction sites |
| air electrode | oxygen pressure at the interface lifts it off |
| steel, air side | chromium vapour deposits and blocks sites |

THE TRADE, SCHEMATIC




cent/h: hotter is cheaper

hours left: fewer

product: an optimum inside

**so the answer** a trajectory $V(t), T(t)$

# Operating for Cost and Lifetime

**Operating strategy** — the cell voltage $V(t)$ and temperature $T(t)$ chosen over the life of the stack to minimise the cost of each kilogram of hydrogen: the electricity bought hour by hour, and the stack replaced when its resistance reaches the end-of-life limit.

## THE LEVELISED COST OF HYDROGEN

$$\mathrm{LCOH} = \frac{C_{stack} + \sum_t p_{el}(t)\, P(t)\, \Delta t}{\sum_t \dot{m}_{H_2}(t)\, \Delta t} \quad \text{over the stack's life}$$

LCOH the cost of a kilogram of hydrogen, $C_{stack}$ the stack, replaced at end of life, $p_{el}(t)$ the hourly price $P(t)$, $\dot{m}_{H_2}(t)$ power and hydrogen at the chosen $V(t), T(t)$; the life ends when $ASR$ reaches $ASR_{\max}$

## THE TRADE-OFFS

| | |
|---|---|
| **higher voltage** | more hydrogen per hour, faster ageing |
| **higher temperature** | lower resistance, faster ageing |
| **cheap hours** | run harder when the price is low |
| **near $V_{tn}$** | small heat, small gradients in the ceramic |

## WHAT THE PINN ADDS

| | |
|---|---|
| **one hour** | the cost in a millisecond |
| **gradient** | $dC/dV$ and $dC/dT$ by autograd |
| **a solver** | one solve per variable per step |
| **a lifetime** | an optimiser over $V(t), T(t)$ |
| **Ex_10.2** | seconds, against a sweep of 81 solves |

```
C = lcoh(net, V_t, T_t)           # cost over a life
V_t = V_t - lr * grad(C, V_t)[0]  # schedule, downhill
```

*Degradation rates are uncertain tenfold: a predicted lifetime ranks strategies, it is not a promise.*

# Advanced Topics

## RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

| topic | what it adds | read | where it makes a difference |
|---|---|---|---|
| across the channel | the gas profile, the porous electrode | Incropera ch. 14 | the 1-D model: +5 – 9 % current; MP10.2A |
| thermal transients | inertia between operating points | Incropera ch. 5 | cycling, start-up, quasi-steady here |
| degradation records | $k_0$, $E_d$, $n$ fitted, not assumed | Liu ch. 5 | lifetime; MP10.2B |
| button-cell fit | kinetics from measured curves | Incropera ch. 1 | validation |
| daily schedule | the hourly price as driver | Liu ch. 5 | grid operation; MP10.2B, L12 |

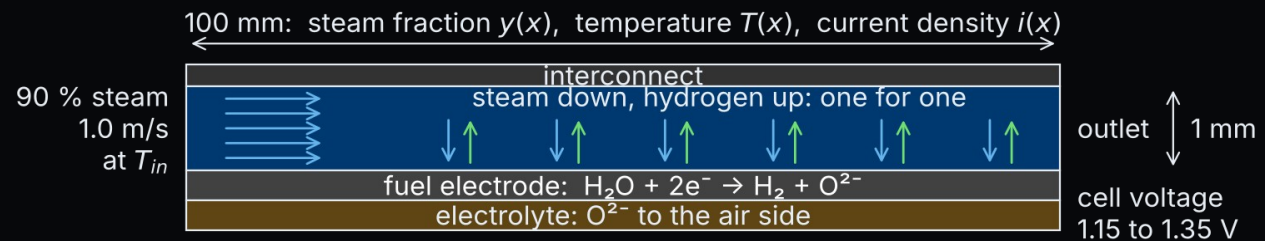
### WHY THEY ARE LEFT OUT

Ex_10.2 solves the coupled channel in one dimension, each operating point at its steady state, scored against a boundary-value solver, so none of them is needed. The first and third rows are the two mini projects; the last meets the grid of L12.

# Exercise 10.2

## ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the electrolyser, its data, the three coupled laws
- **4** — the reference; finite volumes coupled by Newton-Raphson
- **5** — Part A: one PINN for every operating point; 1000 points timed
- **6** — Part B: the optimal voltage under the 10 K limit, by gradient
- **7 – 9** — the report; mini projects MP10.2A and MP10.2B




## THE QUESTION EACH PART ANSWERS

| | |
|---|---|
| **1 – 3** | which laws couple, and how? |
| **4** | what is the coupling worth? |
| **5** | one network, every point? |
| **6** | which voltage, within 10 K? |

## REPORT QUESTION, TO CONCLUDE

For which task in this notebook would you use finite volumes, for which the network - and what changes when a voltage must be chosen for every hour of a day? Say which of your numbers rest on estimates.

SUMMARY

# Key Takeaways

| SOLID OXIDE CELL | |
|---|---|
| **what it is** | a ceramic cell, two directions |
| **function** | steam to hydrogen, heat helping |
| **train, tune** | operated at or near $V_{tn}$ |
| **vs low temperature** | a quarter less electricity |
| **used for** | storing surplus power as fuel |

| LIFETIME OPTIMISATION | |
|---|---|
| **what it is** | a PINN over voltage and inlet heat |
| **function** | three fields, the cost, its gradient |
| **train, tune** | once: 1.15 – 1.35 V, 750 – 850 °C |
| **vs a solver** | autograd gradient; a solver re-solves |
| **used for** | many points, and the choice of $V$ |

*Next — L11.1: a network in a real-time loop, where there is no PDE at all.*

# Questions

## 1 – 2 · A NETWORK FOR THE FUEL CHANNEL

**1** **How can one network learn the steam fraction, temperature and current density along the channel at every operating point?**

The network takes the position along the channel, the cell voltage and the inlet temperature, and returns the three fields. It is trained over 1.15 to 1.35 volts and 750 to 850 degrees until the three laws hold at points over the channel and the operating range; the inlet values are built into the form of the solution, and the current is kept below the limiting current. Any operating point is then read off, not solved again. *Liu ch. 2.2; Ex_10.2 section 5*

**2** **Why does the local current make the channel harder to solve, and how is the network's answer checked?**

The current has no equation of its own along the channel: at every point it is the root of the charge transfer at the local steam fraction and temperature, so no field can be solved first. The network takes it as a third residual, in units of 50 millivolts so it weighs like the balances. Scored against a boundary-value solver at twelve operating points, it is within 0.0022 of the steam fraction and 0.19 kelvin, three times closer than finite volumes on 50 cells. *Ex_10.2 sections 3 – 5*

## 3 – 4 · THE OPTIMAL VOLTAGE

**3** **How can the optimal cell voltage be found with a network, and what does the gradient save?**

The cost of an hour, the electricity bought less the hydrogen sold plus the share of the stack aged, is read from the network's fields, with the 10 kelvin limit as a penalty. Autograd through the network gives its derivative in the voltage, and gradient descent finds 1.250 volts in seconds; without the limit, 1.167 volts, where the cell cools 25 kelvin. A sweep of 81 solver runs agrees. The prices and the ageing law are assumptions, so the optimum is conditional on them. *Liu ch. 5; Ex_10.2 section 6*

**4** **Why does a network help to optimise the operation of the cell over its lifetime, and what should not be claimed from it?**

Over a lifetime the decision is a trajectory of voltage and temperature: what produces more now ages the cell faster. A solver gives the cost, but its gradient costs one solve per variable per step, or an adjoint built by hand; a trained network gives every gradient by automatic differentiation. The degradation constants here are assumed, and published rates span an order of magnitude, so a predicted lifetime ranks schedules; it is not a number to promise. *Liu ch. 5*

NEXT

# L11.1 — Vision-Based Navigation

A network in a real-time control loop on a robot car, where the model is learned from camera frames and there is no PDE at all.

## BEFORE THE NEXT SESSION

| | |
|---|---|
| **read** | Liu ch. 5 |
| **run** | Ex_10.2, sections 1 to 6 |
| **bring** | your optimal voltage, and its worth |
| **think about** | what was physics, what was price |

*Block ten is finished: lithium in a particle, and steam in a ceramic cell - the course's last partial differential equations.*